

Evaluating Classical Image Processing for Brain Tumor Segmentation in MRI Scans

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Github: <https://github.com/ahmedi64/BMEN509>

Introduction

Brain tumors, or intracranial tumors, are abnormal growths of cells within the brain that can significantly impact cognitive and motor functions. These tumors arise due to genetic malfunctions that disrupt normal cell cycles, leading to unchecked cellular growth. Among the most common types are gliomas, meningiomas, and pituitary tumors. Accurately segmenting brain tumors in MRI scans is essential for diagnosis, treatment planning, and monitoring.

This project is aimed to evaluate classical and deep learning-based image processing techniques for segmenting brain tumors from MRI scans. Our approach explored both intensity-based thresholding and advanced convolutional neural networks to determine the most effective segmentation method. Specifically, we investigated Otsu's thresholding, and U-Net, a convolutional neural network architecture commonly used in biomedical image segmentation.

U-Net operates by combining an encoder that captures the context of the image and a decoder that enables precise localization, forming a U-shaped architecture that is effective in creating binary masks of tumor regions. This method enabled us to create accurate segmentation masks for a variety of tumor types.

Dataset

We used the SCIDB Brain Tumour Dataset, which contains over 3000 grayscale MRI images from 233 patients diagnosed with glioma, meningioma, or pituitary tumors. Each image is accompanied by a ground truth segmentation mask, enabling supervised training and evaluation. Although we initially considered using the CBICA dataset, SCIDB was ultimately selected due to its better alignment with our project goals. The dataset can be retrieved from the link below:

<https://www.scidb.cn/en/detail?dataSetId=faa44e0a12da4c11aeee91cc3c8ac11e>

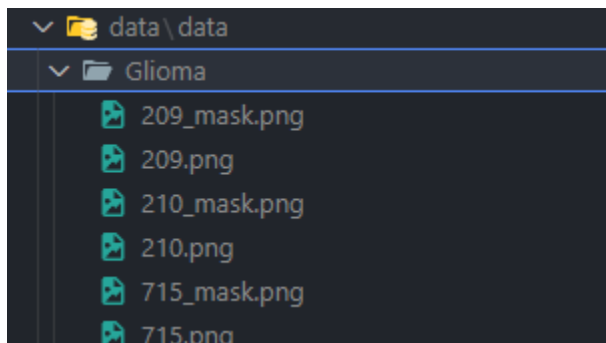
Tools and Frameworks

- Python 3.10
- OpenCV 4.9.0
- TensorFlow 2.15.0
- Matplotlib 3.8.2

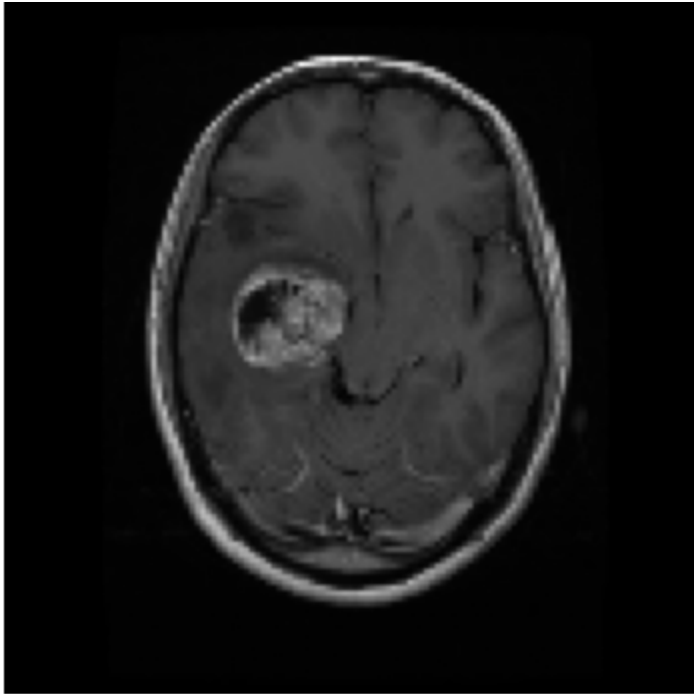
- NumPy 1.26.4
- Scikit-learn 1.4.1

Image Processing Pipeline

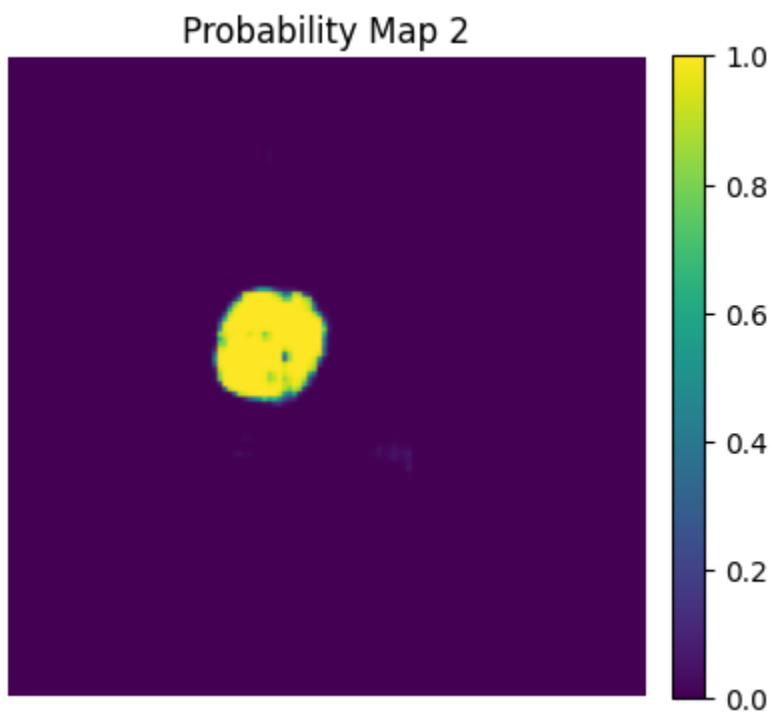
1. **Image Loading:** Select file of MRI scans and corresponding masks. Specifically select the Glioma folder for our model. Load the images in greyscale.



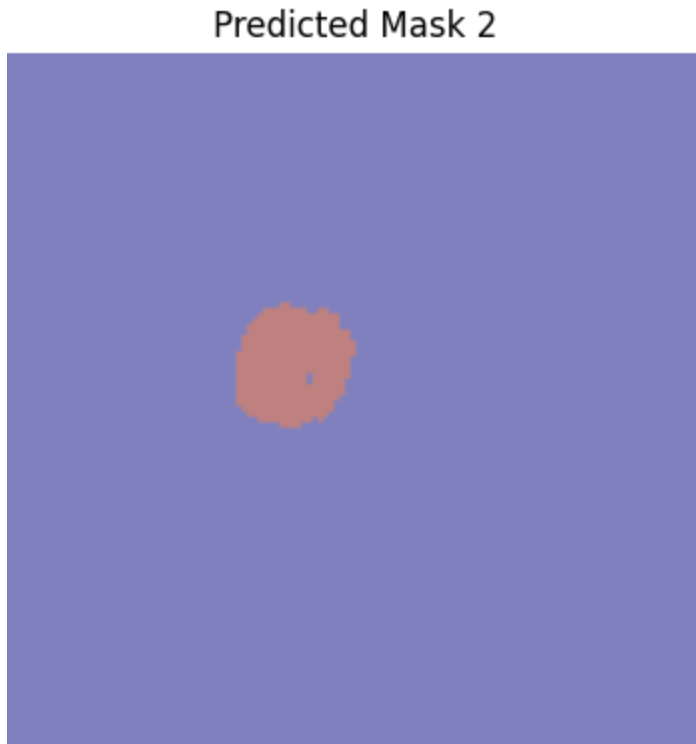
2. **Image Preprocessing:** Resize images to 128x128 pixels, and normalize pixel values between [0,1]. Add an extra dimension for number of channels, which leaves us with the final dimensions to be (128,128,1)



3. **U-net Model:** Pass preprocessed image into trained model. Outputs a probability map for each pixel on the probability that it is a part of the tumor..

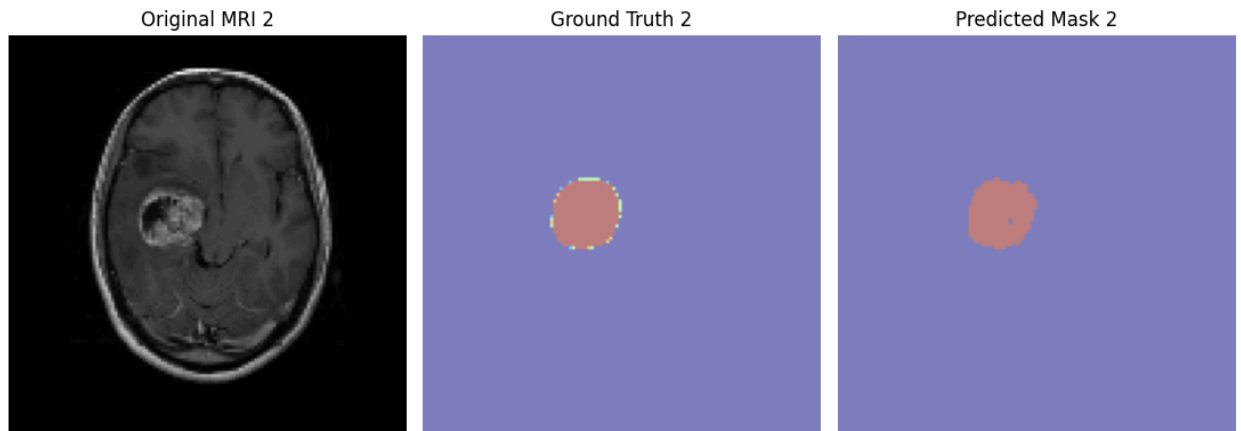


4. **Post Processing:** Apply threshold of 0.5 for probability. Over 0.5 if it is a pixel, and if it is below 0.5, it is not. Create a binary mask based on this threshold.(1 for tumor, 0 for non-tumor). Also remove the extra dimension added before.

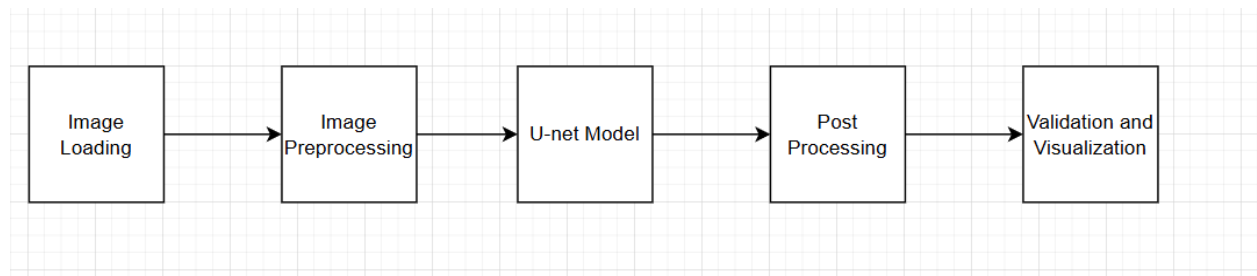


5. **Validate and visualize:** Compare mask to mask given from data set to calculate dice score, and jaccard index. Also print image to user to show user predicted mask

```
1/1 ————— 0s 24ms/step  
Example 2: DSC = 0.7501, IoU = 0.6001
```

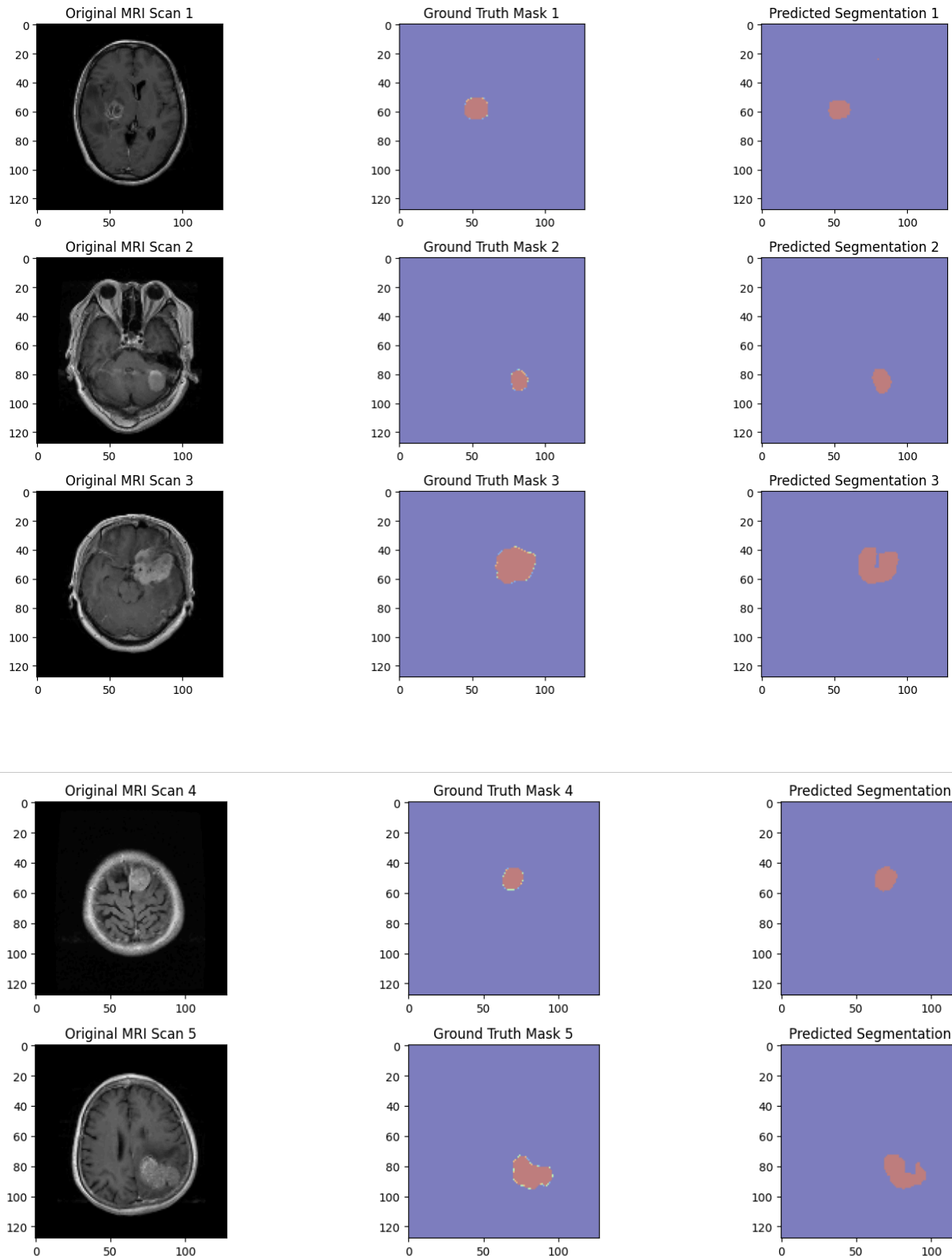


Pipeline Diagram



Results:

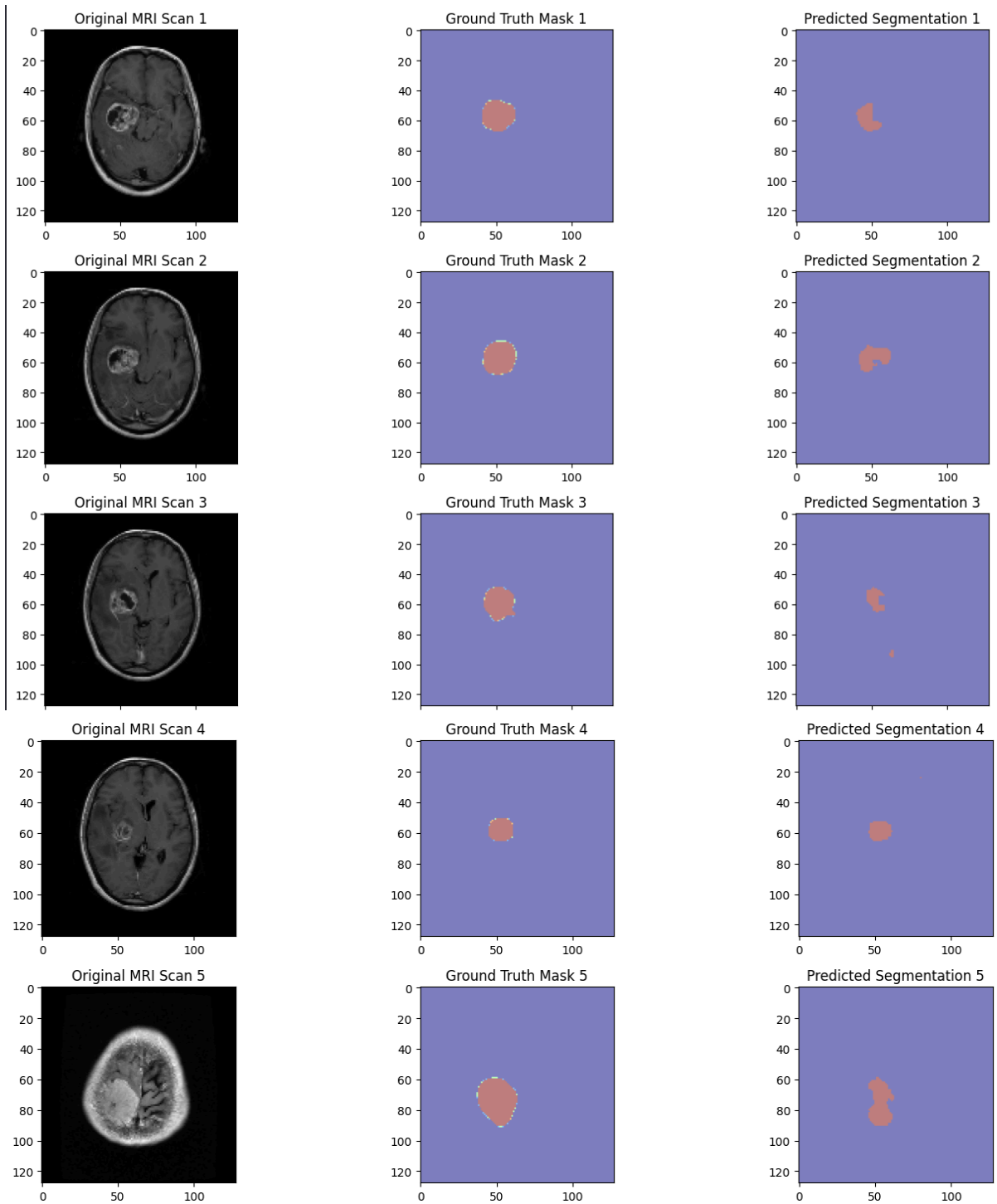
We will present 5 images that we send through the model



The dice scores and Jaccard Index for our 5 above images are:

Example	DSC	IoU	Accuracy
1	0.8700	0.7699	0.9967
2	0.8656	0.7631	0.9971
3	0.9159	0.8449	0.9939
4	0.9328	0.8741	0.9980
5	0.8620	0.7575	0.9924

As seen in the image above, both the dice scores and the Jaccard coefficients are high. This means our model works well for glioma tumors as these 2 parameters are our measure of success. We can also see the predicted masks compared to the truth masks. Visually analyzing it we can see that all our predicted masks are extremely similar to the truth masks, which is reflected in our high dice and IOU scores. Notice also that all these images have very well defined tumors that are evenly white throughout. This brings us to our limitations where again we will show 5 other image processing results.



The dice scores and jaccard score for our above 5 images are:

```
1/1 ----- 0s 108ms/step  
Example 1: DSC = 0.6500, IoU = 0.4815  
1/1 ----- 0s 24ms/step  
Example 2: DSC = 0.7501, IoU = 0.6001  
1/1 ----- 0s 26ms/step  
Example 3: DSC = 0.5072, IoU = 0.3398  
1/1 ----- 0s 24ms/step  
Example 4: DSC = 0.8700, IoU = 0.7699  
1/1 ----- 0s 25ms/step  
Example 5: DSC = 0.7171, IoU = 0.5589
```

As you can see, when our model is being used in cases where the tumor has a dark core, it can cause issues with our model as our model does not recognize it as part of the tumor. This can be seen in examples 1-4. Another issue is when the tumor is especially large, due to it being so large, it confuses the model as most images of tumors are small relative to the brain. As such when an extreme case such as example 5 comes up, it can not fully predict these tumors. All these limitations can be seen in the dice scores and IOU, where they are still decent, however they are nowhere close to the first 5 examples shown before.

Our overall system dice score and jaccard index score are 0.9 and 0.83 respectively. These are respectable scores as the max value for both these parameters are 1. With 1 being a perfect model. This indicates to us that our model is capable of identifying a tumor almost all the time, except when extreme cases occur.

```
Overall Dice Score (F1): 0.9074  
Overall Jaccard Index (IoU): 0.8306
```

Discussion

Our main objective was to develop an image processing pipeline capable of accurately segmenting brain tumours for our dataset of MRI scans, through a high accuracy. Through iterative testing of deep learning techniques, we were able to build a complete segmentation system that aligns with our goal.

The results demonstrated a strong performance in glioma segmentation cases, with an overall **Dice score of 0.9** and **Jaccard Index of 0.83** across our top-performing model. These metrics indicate that our model correctly identifies most tumor regions with high accuracy. Through observations, we further confirmed that in many cases, our model produced highly accurate segmentation masks. This highlights the strength of architectures like U-Net when paired with well-processed data.

However, our pipeline also had some key limitations. Our very first method of Otsu's thresholding failed to capture the glioma tumour boundaries, and lead to over-segmentation in many cases. Even with deep learning, our model struggled with dark-core tumours and unusually large tumours, which could be a limitation with our data as well. In the case that we had a lot more extreme types of tumours to also train our model, it could've helped make our model more robust.

One potential improvement for future work is to implement some data augmentation techniques within the pipeline such as brightness adjustment, or rotations. These methods can expose the model to a wider range of scenarios, and improve its performance on unseen data. Another valuable step would be to expand the dataset to include more tumours with diverse characteristics, like abnormal sizes, shapes and cell distributions. Lastly, while our current pipeline focused primarily on glioma tumours, extending to other types of brain tumours like meningioma and pituitary tumours could also provide a better understanding of the model's effectiveness.

Conclusion

In this project, we developed and evaluated an image processing pipeline to segment brain tumors from MRI scans using both classical and deep learning techniques. Otsu's thresholding provided a simple baseline but suffered from over-segmentation, yielding low Dice scores and limited clinical usefulness. Our first U-Net model achieved a high training and validation accuracy of ~98%, yet the Dice Similarity Coefficient (DSC) was only 0.2734, highlighting the model's struggle to generalize despite strong training performance. After optimizing our preprocessing steps and resizing the input images to 128×128, our final U-Net model demonstrated significant improvement, with an IOU score of 0.4664 and a Dice score of 0.6357, indicating much better overlap with ground truth tumor masks. For future work, we recommend exploring architectures like Attention U-Net to further enhance segmentation accuracy, incorporating a preliminary classification step to identify whether a tumor is present, and deploying the model in a clinical decision-support system to aid radiologists in diagnosis.

References

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